Digital Electronics (ECT-033) - Unit 1

Unit 1 - Minimization of Logic Functions - notes in easy English, matched to the UTU syllabus.

Study Materials by Anas - study-materials-by-anas.vercel.app

What this unit covers: simplifying Boolean functions so the final circuit uses fewer gates - K-maps, Quine-McCluskey, and number-system basics.

Key ideas: Boolean algebra rules (De Morgan, absorption, consensus) turn long expressions into short ones. A **Karnaugh map** (K-map) does it visually: place minterms in Gray-code order, then group 1s in powers of 2 (2, 4, 8...) - each group drops the variables that change inside it. Groups can overlap and wrap around the edges; **don't cares** (X) may join groups but never need their own. SOP grouping gives AND-OR; grouping 0s gives POS. **Quine-McCluskey** is the tabular method for 5+ variables: pair minterms differing in one bit, find prime implicants, then cover the minterms with essential prime implicants. Revise number systems too: binary/hex/octal conversion and 1's/2's complement subtraction.

Exam pointers: one 3-variable and one 4-variable K-map minimisation are near-certain - show groups clearly on the map. A small Quine-McCluskey table (given minterm list) and a "convert + complement arithmetic" question round out the unit.

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